

Total No. of Questions : 10]

SEAT No. :

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[5669]-114

T.E. (Computer Engineering)

DIGITAL SIGNAL PROCESSING APPLICATIONS

(2012 Pattern) (Semester - II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7, or Q.8, Q.9. or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data if necessary.

- Q1)** a) Classify DT systems as FIR and IIR systems. Express it by means of suitable mathematical form using convolution and difference equation. [5]
- b) How DFT is different than Fourier Transform (FT)? How one can plot the magnitude spectrum of DFT? [5]

OR

- Q2)** a) State the sampling theorem and explain why the problem of aliasing observed during sampling process? [5]
- b) Obtain the ZT of [5]

$$x(n) = \left(\frac{1}{2}\right)^n u(n) \text{ sketch the ROC}$$

- Q3)** a) Derive the first stage of Radix-2 DIT FFT algorithm. [5]
- b) Determine the inverse Z-transform using partial fraction expansion method. [5]

$$H(Z) = \frac{1}{1 - 1.5Z^{-1} - 0.5Z^{-2}} \text{ if}$$

$$\text{ROC: } |Z| > 1$$

P.T.O.

OR

Q4) a) Compute 4 point Circular Convolution for DT signals. [5]

$$x_1(n) = \{1, 2, 3, 1\} \quad x_2(n) = \{4, 3, 2, 2\}$$

b) State & Prove the Time Shifting and Time Reversal properties of Fourier Transform. [5]

Q5) a) What are filter structures? How the Direct form and Cascade form of FIR filters are obtained and realized? [9]

b) Realize the system described by following difference equation using direct form -I [9]

$$y(n) = y(n-1)^{-1/2} y(n-2) + x(n) - x(n-1) + x(n-2)$$

OR

Q6) a) Obtain and realize Linear Phase FIR filter structure for a DT system. [9]

$$y(n) = x(n) + \frac{1}{3}x(n-1) + \frac{1}{4}x(n-2) + \frac{1}{4}x(n-3) + \frac{1}{3}x(n-4) + x(n-5)$$

What are the advantages of this filter structure?

b) Derive the Direct Form-II IIR filter structure from system function $H(Z)$ and represent it using multipliers, adders and delay elements. [9]

Q7) a) Explain the features of SHARC DSP processor. List the number of DAGs with its capabilities and memory pointer registers supported by DAG. [8]

b) Explain applications of DSP with respect to following [8]

i) Telecommunication

ii) Biomedical

OR

Q8) a) Explain and compare following architectures with suitable block diagram. [8]

i) Von Neumann Architecture

ii) Harvard Architecture

iii) Modified Harvard Architecture

b) Draw and explain the SIMD (Single Instruction Multiple Data) architecture of SHARC DSP processor [8]

- Q9) a)** Draw and explain Human Speech Model in speech synthesis and recognition. **[8]**
- b)** How digital image is represented by means of digital computer? How gray scale image is different than color image? What is Histogram of an image? **[8]**

OR

- Q10)a)** What is Companding? What is its significance in audio processing? What is the impact of data rate on sound quality? **[8]**
- b)** With mathematical form, explain any two gray level transforms used for image enhancement. **[8]**

